Design Document (M2): Gecko Enclosure Humidity Advisor

CS 370 Term Project · Authors: Gerritt Ewing, Zoe Ebert · Status: draft. Items marked **[Pi: unverified]** still need real evidence from the Pi. Items marked **[proposed]** await confirmation from both partners.

User: Rosalie Endris, keeper of a gecko. The device is advisory only: it senses, logs, analyzes and displays on a LAN dashboard. It never controls the humidifier, and it works with the network cable unplugged. Two sensors cooperate in one pipeline: a DHT11 (temperature and humidity, GPIO) and a TSL2591 (light, I2C). They run at different rates and interfaces, and the hub reconciles them into one fused snapshot.

1. Architecture

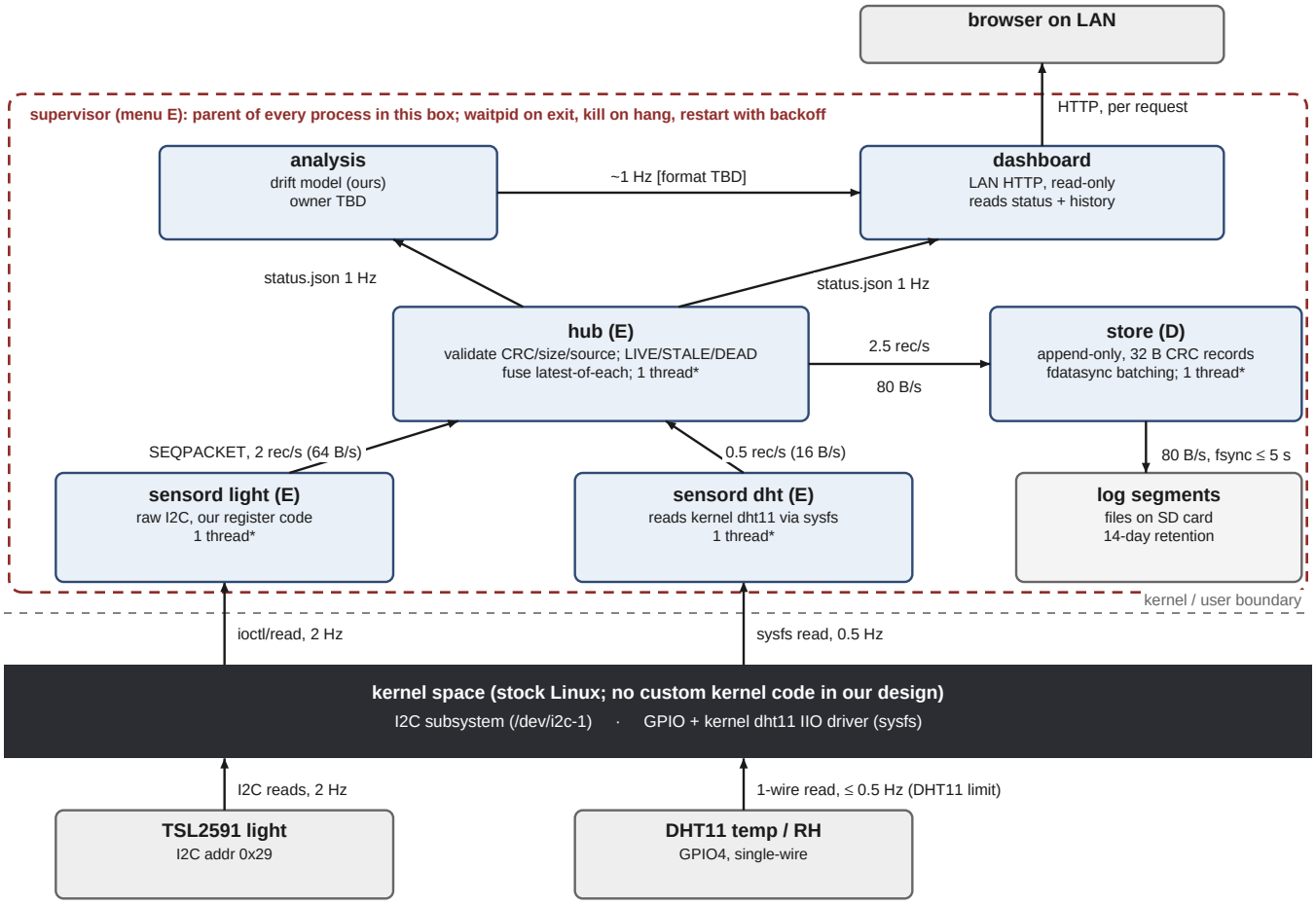

Figure 1. Processes, kernel components, data flows and rates. Letters mark the menu mechanisms (E, D). Every process is a separate C binary under `src/`. *Threads: each process is a single-threaded event loop and concurrency comes from separate processes **[proposed]**. Kernel components are stock (I2C subsystem, kernel `dht11` IIO driver); our code starts at the sensord boundary. The dashboard also reads history from the log on request. Rates are targets until measured **[Pi: unverified]**; the DHT11 cannot be read faster than about 1 Hz and about 10% of reads fail.

IPC contract (sensord, hub, store).

- Transport: `AF_UNIX`, `SOCK_SEQPACKET`. Each message is exactly one 32-byte `struct cs370_record` (`src/common/record.h`). SEQPACKET keeps message boundaries, and a dead peer shows up as EOF on the next read, so the hub learns about a dead sensord without waiting for a timeout.
- Sockets: the hub listens on `$CS370_RUN_DIR/hub.sock` and every sensord connects to it. The store listens on `$CS370_RUN_DIR/store.sock` and the hub connects to it. `CS370_RUN_DIR` defaults to `$XDG_RUNTIME_DIR/cs370`.
- Every fd is opened `CLOEXEC`, so a restarted child never inherits a stale socket. Every send uses `MSG_NOSIGNAL`, so a dead peer returns `EPIPE` instead of killing the sender with SIGPIPE.
- sensord does not buffer. If the hub is down it keeps sampling, drops and counts the samples, and reconnects with exponential backoff (0.5 s, doubling up to 8 s). The gap shows in the store as a `seq` jump.
- The hub forwards every valid record to the store. While the store is down it holds up to 256 records (about 100 s), then drops the oldest, counted and logged. Invalid records (wrong size, bad CRC, unknown source) are rejected and logged, never forwarded.

Liveness (hub), per source. Thresholds are targets until measured **[Pi: unverified]**.

State	Condition
LIVE	a valid, non-MISSING sample within 3 sample periods (DHT 6 s, light 1.5 s)
STALE	no valid sample for 3 periods, or the sensord connection closed
DEAD	no valid sample for 60 s
SUSPECT	the latest value failed the range or rate-of-change check (tracked separately from staleness)

Every second the hub atomically replaces `$CS370_RUN_DIR/status.json` (temp file, then `rename`). For each source it carries the state, last value, age and FAKE flag. The dashboard renders this state and never infers freshness itself.

2. Mechanism mapping

Menu item	Component	Justification from Rosalie's needs
E multi-process + supervisor	`sensord-*`, `hub`, `supervisor`	A dead sensor must never look like "humidity is fine". Each sensor runs in its own process, so a hung DHT11 read cannot stall light sampling or the dashboard. When a sensord dies, its connection hits EOF and the hub marks that source STALE immediately; after 60 s of silence it is DEAD. The supervisor restarts it and the gap is visible in the log as a `seq` jump. The dashboard shows STALE or DEAD instead of the last value.
D crash-consistent storage	`store`	The drift model needs days of history, and Rosalie's question after a bad night is "what happened", so the log must survive a power cut. Append-only fixed-size CRC records plus a recovery scan that truncates a torn tail.

Not chosen, and why. A (kernel driver) adds no value for a slow sensor. B and C do not matter at DHT11 timescales: a few ms of jitter is irrelevant to humidity that moves over minutes. F has no rate pressure. We say so plainly rather than bolt on a mechanism. Both chosen mechanisms are written in C by us and are measured in the evaluation (section 6).

3. Failure-mode table

Component	Failure	Detection	System response	Log shows
sensord-dht	single read fails (checksum/timeout, ~10%)	read returns error	retry once after 2 s; report miss, no value	`dht miss n=<consecutive>`
sensord-dht	sensor unplugged / dead	N consecutive misses or no sample for 60 s	mark DEAD; dashboard banner; other sensor keeps running	`dht DEAD since <ts>`
sensord-light	I2C NACK / bus error	errno from ioctl/read	reinit device; after N failures mark DEAD	`light err errno=<n>`
any sensord	process killed (`kill -9`)	supervisor `waitpid`	restart with backoff, new generation id	`child dht died sig=9 restart=<n>`
any sensord	hang (alive, no output)	no heartbeat/sample within deadline	supervisor SIGKILL then restart	`child dht hung, killed`
hub	exits	supervisor; sensords get `EPIPE`/`ECONNREFUSED`	restart; sensords count dropped samples and reconnect (0.5 s up to 8 s)	`hub restart`, `sensord dht hub lost, dropped=<n>`
hub	store down / not yet started	`connect` fails or `EPIPE` on send	hold up to 256 records, reconnect with backoff, drop oldest on overflow	`hub store lost`, `hub store drop n=<total>`
hub	malformed record from a client	size != 32, CRC mismatch or unknown source	reject the record, keep the connection	`hub reject src=<fd> reason=crc`
sensord	dht11 driver/overlay missing	sysfs path absent at startup	exit 1 (supervisor logs and backs off); dashboard shows DHT DEAD	`sensord dht no iio device`
store	power cut mid-write	CRC/length check at startup scan	truncate to last good record; continue	`recover truncated <bytes>`
store	disk full	`write` returns ENOSPC	stop appending, dashboard flag, keep sensing	`store ENOSPC`
supervisor	dies	systemd (or runner) restarts it; children detect parent loss and exit	clean restart; orphan fds closed	`supervisor start gen=<n>`
clock	jump (no RTC, NTP step)	monotonic vs wall delta	order by monotonic + seq; wall time is advisory	`clock step <s>`
any	sensor value implausible	range / rate-of-change check	mark SUSPECT, do not feed model	`dht suspect rh=<v>`

Rule from CLAUDE.md: stale, dead or implausible data is shown as such, never as the last good value. The log strings are the planned formats; the C daemons are not yet implemented.

4. Storage and data

Record: fixed 32 bytes, little-endian, no padding, defined in `src/common/record.h` (the header is the source of truth; `tests/c/test_record.c` checks every offset).

Off	Field	Type	Meaning
0	`mono_ns`	u64	`CLOCK_MONOTONIC` ns: ordering and duplicate detection
8	`seq`	u32	per-source sequence number
12	`wall_s`	u32	unix seconds, advisory only (the clock can jump)

Off	Field	Type	Meaning
16	`lux`	f32	TSL2591 lux; 0 unless source = light and not MISSING
20	`rh`	u16	%RH x10; 0 unless source = dht and not MISSING
22	`temp_c`	i16	degC x10; 0 unless source = dht and not MISSING
24	`source`	u8	0 = dht, 1 = light
25	`flags`	u8	bit 0 FAKE, bit 1 SUSPECT, bit 2 MISSING; 0 = real
26	`reserved`	u8[2]	must be zero
28	`crc32`	u32	IEEE CRC-32 of bytes 0-27

- A failed DHT read is a MISSING record (value fields zero), never a zero reading. On the Pi, the DHT11 reported temperature in 0.1 degC steps and humidity in whole %RH, so `rh` is a multiple of 10 today.
- Valid record (`cs370_record_valid()`, used by hub and store): CRC matches, `source < 2`, no undefined flag bits, `reserved` zero, `lux` finite and >= 0, and fields that do not apply to the source are zero.
- SUSPECT bounds (targets **[Pi: unverified]**): DHT11 temperature outside 0-50 degC or RH outside 20-90 %, or a change of more than 5 degC or 20 %RH between consecutive samples; light outside 0-88000 lux or a jump not explained by a gain change.
- Rate: 2 Hz light + 0.5 Hz DHT = 2.5 records/s; 32 B x 2.5 = 80 B/s, about 6.9 MB/day.
- Retention: 14 days (about 100 MB), rotated by segment file with the oldest deleted; the dashboard reads history from the log, not from memory.
- Durability: `write` per record into a buffered segment, `fdatasync` every N records or 5 s, whichever comes first. The loss window is at most 5 s and will be reported as a measured number.
- Power dies mid-write: on start, scan the tail segment, verify each CRC, and truncate at the first bad record.
- Replayed or synthetic records carry the FAKE flag, and the dashboard and logs label them.

5. Constraints and substitutions

Ideal	Actual	Cost
SHT45 RH/temp (+/-1% RH, 0.1 C)	DHT11 (about +/-5% RH, whole-percent RH, 0.1 C steps seen, 1 Hz, ~10% missed reads [Pi: unverified])	cannot resolve small drifts; the model works on trends over minutes and claims are narrowed accordingly
DS18B20 second temperature	none	no independent temperature cross-check; the light sensor is the second source
Actuator-in-the-loop control	advisory only	no closed-loop result to measure; evaluation is prediction quality, not control quality
Months of enclosure data	days of provoked episodes	claims limited to episodes we induce (lamp on/off, vent open, misting)
DHT11 read with timing we control	kernel `dht11` IIO driver (`dtoverlay=dht11,gpiopin=4`) via sysfs	the bit timing is the kernel's, not ours (not one of our chosen mechanisms). Bit-banging 26-70 us pulses from userspace is where scheduling jitter breaks reads. The overlay claims GPIO4, so Python `adafruit_dht` no longer works on the Pi. Fallback: libgpiod v2 edge timestamps

Pi evidence (2026-10-27, piewing, kernel 6.18.50+rpt-rpi-v8 aarch64, libgpiod 2.2.1).

- TSL2591 is alive, checked with raw `i2c-tools` using the same register addresses the C backend uses: `i2cdetect -y 1` shows 0x29 and nothing else; the ID register (0xB2) reads 0x50; after ENABLE = PON|AEN the AVALID bit is set; channel counts were near 0 at medium gain in a dark room and steady, non-zero at maximum gain, and scale with the gain setting.
- On this bus SMBus word reads return only one byte per register, so the C backend reads C0DATAL..C1DATAH as one 4-byte block starting at command 0xB4.
- Lux: with C0 (visible + IR) and C1 (IR), `CPL = ATIME_ms * G / 408.0` (G: medium 25x, max 9876x) and `lux = max((C0 - C1) * (1 - C1/C0), 0) / CPL`. A saturated channel (36863 counts at 100 ms, 65535 at longer times) is reported SUSPECT. Gain and integration time are fixed at startup so lux stays comparable.
- **DHT11, alive (2026-10-28).** With `dtoverlay=dht11,gpiopin=4` added under `[all]` in `/boot/firmware/config.txt` and a reboot, the kernel exposes `/sys/bus/iio/devices/iio:device0` named `dht11@4` (the name carries the GPIO number, so sensord must match the `dht11` prefix, not the exact name). 20 reads, 2 s apart, returned `in_temp_input` = 17800 to 18300 and `in_humidityrelative_input` = 39000 to 47000. These are milli-units (17.8 to 18.3 degC, 39 to 47 %RH), so dividing by 100 gives the record's x10 fields. Temperature moved in 0.1 degC steps and humidity in whole percent. No read failed in the 20, so the ~10% miss rate is still an unverified estimate **[Pi: unverified]**; a longer run is pending.

6. Evaluation plan

Measurement	Method	Target
Sample-to-log latency distribution	timestamp at sensord read vs at log append; p50/p95/p99, idle and under CPU stress	p99 under 250 ms (to confirm)
Per-process CPU and RSS	heartbeat series over the 48 h soak	RSS flat (no leak), CPU under 10%
Domain metric: band-exit prediction	N provoked episodes labeled by us; drift model vs a plain threshold on lead time, misses and false alarms	model alerts earlier than the threshold with no more false alarms
Fault injection (mechanism E)	`kill -9` each child, unplug each sensor 10 min, power cut mid-write	each row in section 3 produces its log line and recovers
Durability (mechanism D)	cut power at random during writes, repeat M times	zero corrupt records read back; loss window within 5 s

7. Ownership map and AI-use plan

Contributions so far: Gerritt set up the repo, deploy and test scaffolding, the fake sensor code, and wired and configured the sensors on the Pi. Zoe drove the project design, including choosing the two-sensor idea, and wrote the README contributions. Ownership means first authorship and answerability, not exclusivity **[proposed]**.

Subsystem	Owner
sensord-dht, sensord-light	Zoe
hub: the pipeline that fuses both sensors (merge, liveness, status snapshot, forwarding to store)	Gerritt
record format (`src/common/record.h`)	Zoe
store (format, recovery)	Zoe
analysis (drift model)	TBD
supervisor, dashboard	shared

AI-use plan. Claude Code is used for exploring, drafting from approved plans, tests, and adversarial review of diffs. It is not used for the drift model's design decisions, the failure-mode table claims, or anything at runtime. The boundary stays visible in the repository: no network client in `src/` (the only socket is the LAN listener, and a CI grep (planned) will check for `connect(`, curl and HTTP-client usage), and no model files in the tree. Hard constraints live in CLAUDE.md, which both partners maintain.